Solved Example: Neural Network Forward and Backward Propagation

Problem: Consider a simple neural network with:

- Input layer: 2 neurons (x_1, x_2)
- Hidden layer: 2 neurons (h_1, h_2)
- Output layer: 1 neuron (y_{pred})
- Activation: Sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$

Given:

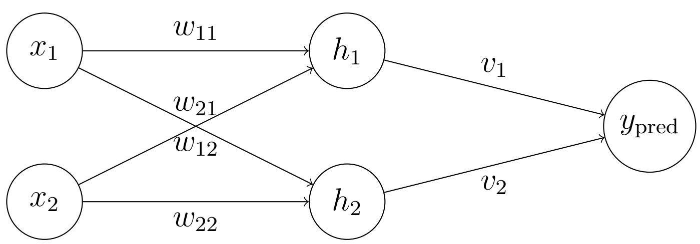
$$x = \begin{bmatrix} 0.05 \\ 0.10 \end{bmatrix}, \quad y_{\text{true}} = 0.01$$

Weights and biases:

$$W_1 = \begin{bmatrix} 0.15 & 0.20 \\ 0.25 & 0.30 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0.35 \\ 0.35 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 0.40 & 0.45 \end{bmatrix}, \quad b_2 = 0.60$$

Step 0: Diagram of Neural Network



Explanation: Each weight in W_1 represents the strength of connection from an input neuron to a hidden neuron. Bias b_1 shifts the activation of the hidden neurons. W_2 and b_2 do the same for the hidden-to-output connections.

Step 1: Forward Propagation

Hidden layer input:

$$z_1 = W_1^T x + b_1 = \begin{bmatrix} 0.15 & 0.25 \\ 0.20 & 0.30 \end{bmatrix} \begin{bmatrix} 0.05 \\ 0.10 \end{bmatrix} + \begin{bmatrix} 0.35 \\ 0.35 \end{bmatrix} = \begin{bmatrix} 0.3825 \\ 0.39 \end{bmatrix}$$

Hidden layer output:

$$a_1 = \sigma(z_1) \approx \begin{bmatrix} 0.594 \\ 0.596 \end{bmatrix}$$

Output layer input:

$$z_2 = W_2 a_1 + b_2 = 0.40(0.594) + 0.45(0.596) + 0.60 \approx 1.1058$$

Output:

$$y_{\text{pred}} = \sigma(z_2) \approx 0.751$$

Step 2: Loss Calculation

Mean Squared Error:

$$L = \frac{1}{2}(y_{\text{true}} - y_{\text{pred}})^2 = \frac{1}{2}(0.01 - 0.751)^2 \approx 0.2745$$

Step 3: Backward Propagation

Output Layer Gradients:

$$\frac{\partial L}{\partial y_{\text{pred}}} = y_{\text{pred}} - y_{\text{true}} = 0.741$$

$$\frac{\partial L}{\partial z_2} = 0.741 \cdot \sigma'(z_2) \approx 0.741 \cdot 0.188 \approx 0.139$$

$$\frac{\partial L}{\partial W_2} = 0.139 \cdot a_1^T \approx \begin{bmatrix} 0.0825 \\ 0.0828 \end{bmatrix}, \quad \frac{\partial L}{\partial b_2} = 0.139$$

Hidden Layer Gradients:

$$\frac{\partial L}{\partial a_1} = W_2^T \cdot 0.139 \approx \begin{bmatrix} 0.0556 \\ 0.0626 \end{bmatrix}$$

$$\frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial a_1} \odot \sigma'(z_1) \approx \begin{bmatrix} 0.0134 \\ 0.0151 \end{bmatrix}$$

$$\frac{\partial L}{\partial W_1} = x \cdot (\frac{\partial L}{\partial z_1})^T \approx \begin{bmatrix} 0.00067 & 0.000755 \\ 0.00134 & 0.00151 \end{bmatrix}, \quad \frac{\partial L}{\partial b_1} = \begin{bmatrix} 0.0134 \\ 0.0151 \end{bmatrix}$$

Step 4: Update Weights (Gradient Descent)

Using learning rate α:

$$W_1^{\text{new}} = W_1 - \alpha \frac{\partial L}{\partial W_1}, \quad b_1^{\text{new}} = b_1 - \alpha \frac{\partial L}{\partial b_1}$$

$$W_2^{\text{new}} = W_2 - \alpha \frac{\partial L}{\partial W_2}, \quad b_2^{\text{new}} = b_2 - \alpha \frac{\partial L}{\partial b_2}$$

Summary: The diagram and weight matrices show how each input contributes to each hidden neuron and eventually to the output. During backpropagation, the gradients tell us how to adjust weights to reduce the error.